# Aryan Bhagat

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## EDUCATION

**California State University, East Bay** **Aug 2024 – May 2026**
*Master of Science - **Computer Science**, GPA - 3.9/4.0* *Hayward, CA, USA*

**L.D. College of Engineering, Gujarat Technological University** **Oct 2020 – May 2024**
*Bachelor of Engineering - **Computer & Electronics Engineering**, CGPA - 8.8/10.0* *Ahmedabad, India*

## WORK EXPERIENCE

**AI Engineer Intern** **Jul 2025 – Sep 2025**
*Simple Ticket* *Boston, MA, USA*

- Engineered **AI-powered CRM chatbot systems** with **OCR-based document ingestion**, semantic retrieval, and **Retrieval-Augmented Generation (RAG)** pipelines using **vector databases, LangChain,** and **LangGraph**.
- Built **API-driven agent workflows** with **RESTful services** and **LangChain Expression Language (LCEL)**, enabling **multi-step reasoning**, tool orchestration, and **stateful execution** for production-grade AI assistants.

**Research Assistant** **Sep 2025 – Mar 2026**
*College of Science, California State University - East Bay* *Hayward, CA, USA*

- Developed real-time robotic perception pipelines on **JetAuto Pro** (**Jetson Orin Nano**) using **ROS2, OpenCV,** and **3D depth cameras** for **gesture recognition, object detection,** and **autonomous navigation**.
- Evaluated **AI-assisted robotic systems** with **AR-based visualization** and **LiDAR-driven spatial mapping**, integrating **sensor fusion techniques** to improve navigation accuracy and reduce end-to-end system latency by **15%**.

**Firmware Engineer** **Jan 2024 – Jun 2024**
*Indiesemic Private Limited* *Ahmedabad, India*

- Developed embedded applications using **Zephyr RTOS** on **Nordic nRF52/nRF53 DKs** and **Quectel EVKs**, performing low-level debugging with **Segger J-Link, RTT,** and **Nordic nRF Connect SDK** to optimize system performance.
- Architected a **BLE-based communication framework** with multi-protocol support **(UART, SPI, I2C, LoRa)**, enabling concurrent peripheral communication and improving overall system throughput by **10%**.

**Teaching Associate** **Jul 2025 – May 2026**
*College of Science, California State University - East Bay* *Hayward, CA, USA*

- Led **C++ Data Structures & OOP** (**CS201**) lab instruction for **30+ students**, covering **ADTs, algorithm design, STL,** and **time-space complexity analysis**.
- Mentored students in **debugging, profiling,** and **performance optimization**, reinforcing software engineering best practices through implementation of efficient **sorting** and **searching algorithms**.

## PROJECTS

**Distributed Event-Driven Notification Platform** [GitHub]

- Architected a distributed notification system using **Apache Kafka, Spring Boot,** and microservices, implementing priority-isolated topics, consumer groups, and concurrent delivery pipelines for Email (SendGrid), SMS (Twilio), and Push notifications.
- Built an event-driven asynchronous processing architecture with retry handling, priority-based routing, and external API orchestration, enabling fault-tolerant, low-latency notification delivery under high-throughput workloads.

**Smart Recipe Tagger – Multimodal AI Recipe Platform** [GitHub]

- Engineered a full-stack AI application using **React, Node.js, MongoDB, Firebase,** and **Google OAuth**, enabling scalable ingestion, management, and retrieval of user food images through cloud-hosted workflows.
- Designed a multimodal AI pipeline integrating **Google Vision API** for ingredient detection and **Gemini API** for recipe and nutrition generation, deployed via **Docker + Google Cloud Run** for scalable low-latency inference.

**FactShield – Ensemble Fake News Detection Platform** [GitHub]

- Built an AI-powered misinformation detection platform using **Flask, RoBERTa, XGBoost, LightGBM,** and **Scikit-learn**, implementing ensemble NLP pipelines for real-time fake news classification.
- Designed a weighted multi-model inference engine combining transformer-based semantic analysis with traditional ML classifiers to improve prediction robustness and confidence scoring across news articles.

**RAG-Based Personalized Outreach Engine** [GitHub]

- Developed a GenAI-powered outreach platform using **Llama 3.3, Groq, LangChain, ChromaDB,** and **Streamlit** to generate context-aware cold emails using retrieval-augmented generation (RAG).
- Built data pipelines for web scraping, semantic embedding, and vector similarity search to retrieve company and role-specific context, improving personalization and grounding LLM-generated outputs.

## SKILLS

**Languages:** Python, C++, Java, JavaScript, TypeScript, Go, SQL
**Backend & Distributed Systems:** Node.js, Express.js, FastAPI, Flask, RESTful APIs, WebSockets, Kafka, Microservices
**Frontend:** React.js, Next.js, Vue.js, HTML5, CSS3, TailwindCSS, Chakra UI, Framer Motion
**AI/ML:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, Hugging Face Transformers, MLflow, Prompt Engineering
**Databases & Cloud:** PostgreSQL, MongoDB, ChromaDB, NoSQL, AWS (EC2, S3), Docker, CI/CD Pipelines, Git, GitHub
**Engineering:** DSA, OOP, Design Patterns, Debugging, Profiling, Performance Optimization, System Design, Agile, Testing